Chapter 3 Quiz Name:

1. A child bikes East on a straight path. The table below shows the child's position (in meters) at a few select times (in seconds).

t time (s)	24	26	124	126
x position (m)	101	109	820	830

a. Estimate the velocity at $t = 25$ s.

b. Estimate the velocity at $t = 125$ s.

c. Estimate the acceleration from $t = 25$ s until $t = 125$ s.

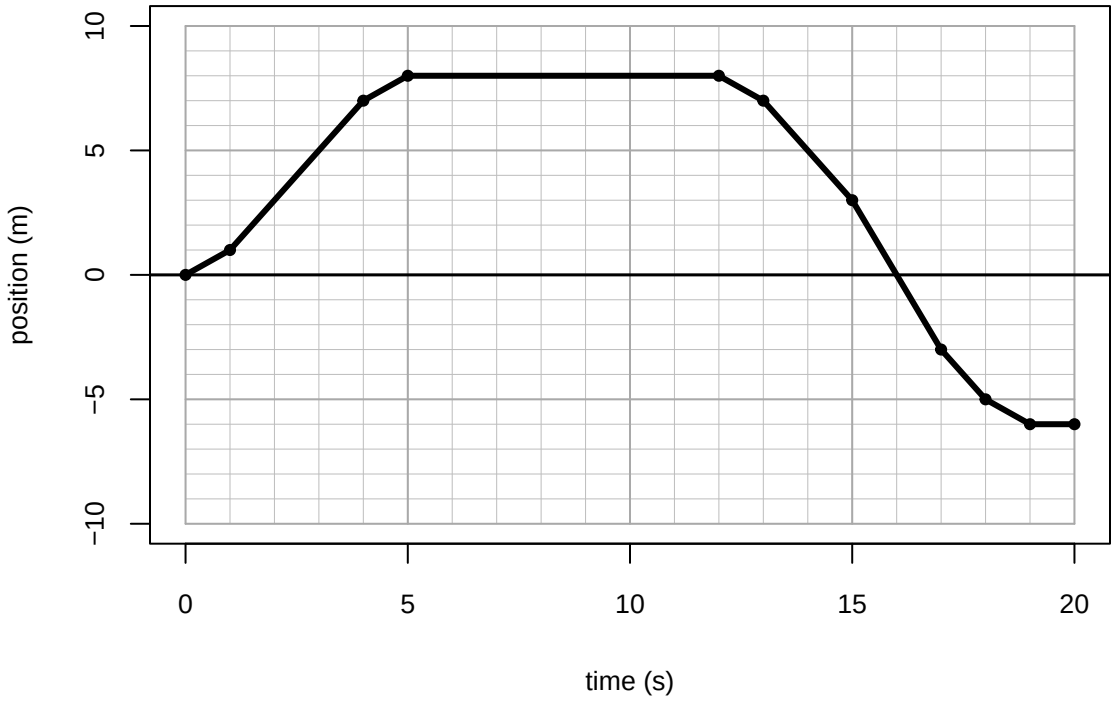
2. A speeding car has a speed of 30 m/s. By slamming the brakes as hard as possible on dry asphalt, the car can decelerate (slow down) at about 10 m/s^2. (Or, if the direction of travel is positive, the braking causes an acceleration of -10 m/s^2.)

a. How much time elapses between slamming on the brakes and the car coming to a stop?

b. While decelerating (in the time between slamming on the brakes and coming to a stop), the average velocity of the car is 15 m/s. Using the time you calculated in part (a), how far did the car travel while decelerating?

c. If the speeding car slams into a tree, it goes from 30 m/s to 0 m/s in 0.05 seconds. Calculate the (average) acceleration experienced during the crash.

3. An elevator begins at ground level. It travels up and down. The elevator's **position** (y in meters) is plotted as a function of time (t in seconds).



a. When $t = 3$, what is the position of the elevator?

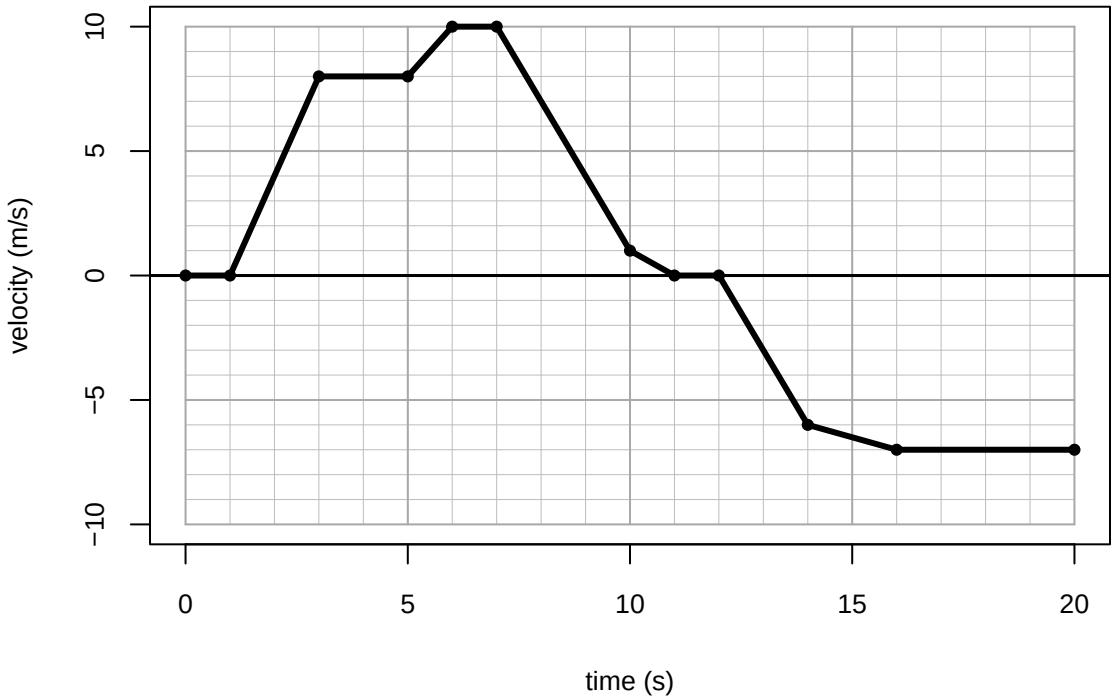
b. When $t = 3$, what is the velocity (speed and direction) of the elevator?

c. During which time interval(s) is the elevator above ground level (when is $y > 0$)?

d. During which time interval(s) is the elevator moving up?

e. During which time interval(s) is the elevator stationary (not moving)?

4. A dog runs North and South along a beach. Let positive velocity indicate the dog is running North.



a. What is the velocity (speed and direction) at $t = 8$ s?

b. What is the acceleration at $t = 8$ s?

c. When is the dog stationary (not moving)?

d. When is the dog moving North?

5. A salmon swims 4 m/s relative to the surrounding water. The water is flowing North at 3 m/s.

a. Relative to ground, what is the velocity of the salmon when the salmon swims **South** at 4 m/s relative to the water? Please draw a diagram of vector addition.

b. Relative to ground, what is the velocity of the salmon when the salmon swims **North** at 4 m/s relative to the water? Please draw a diagram of vector addition.

c. Relative to ground, what is the velocity of the salmon when the salmon swims **East** at 4 m/s relative to the water? Please draw a diagram of vector addition.